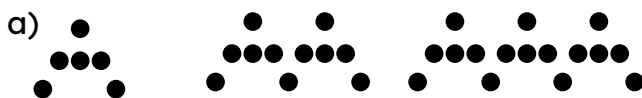




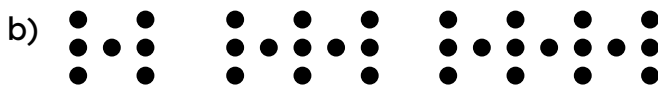
# Generating a sequence using a position-to-term rule

## Task A: Writing position-to-term rules

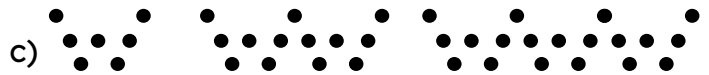
1) For each sequence, write a **position-to-term** rule for the number of dots in each pattern.



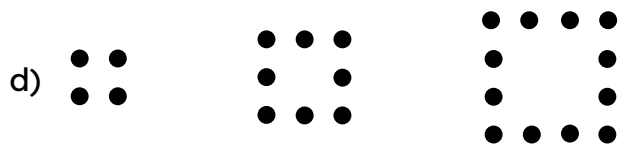
| Term number | 1 | 2 | 3 |
|-------------|---|---|---|
| Term        |   |   |   |



| Term number | 1 | 2 | 3 |
|-------------|---|---|---|
| Term        |   |   |   |



| Term number | 1 | 2 | 3 |
|-------------|---|---|---|
| Term        |   |   |   |



| Term number | 1 | 2 | 3 |
|-------------|---|---|---|
| Term        |   |   |   |

2) Four students are writing a **position-to-term** rule for the sequence:

9, 11, 13, ... Who is correct?

For the incorrect answers, show how you know they are incorrect.

The term is the term number multiplied by 9



Sam

The term is the term number add 2



Sofia

The term is the term number multiplied by 2 add 7



Alex

The term is the term number multiplied by 2 add 9



Izzy



**Task B: Generating terms from position-to-term rules**

For each sequence, find the value of  $T_{60}$

- a) Each term is the term number multiplied by 5 plus 6
- b) Each term is the term number halved then add 2
- c) Each term is the term number subtract 5
- d) Each term is 10 less than the term number then multiplied by 3
- e) 12, 15, 18, 21, .... (with a constant additive pattern)
- f) 18, 19, 20, 21, ... (with a constant additive pattern)
- g) 1.4, 1.8, 2.2, 2.6, ... (with a constant additive pattern)